

Case Presentation

호흡기알레르기내과

PK A-7조

202152517 김민지

Chief Complaint & Present Illness

- Patient : 72Y / F
- **CC** : 내원 4일 전부터 시작된 Dyspnea (mMRC 3)
- **PI** :
 - 2013년 BE 진단 후 호흡기내과 추적관찰 중인 환자로, 내원 4시간 전부터 dyspnea 악화되어 응급실 내원함.
 - 1-2개월에 한 번씩 가래에 손톱보다 소량의 객혈 있었음.
 - 내원 5일 전부터 노란색 가래, 콧물 동반되었고 내원 4일 전부터 DOE 발생하여 계단 1층 오르거나 100 m 보행 시 호흡곤란 호소함. 쉬면 호전되는 양상이었음.
 - 내원 2일 전부터 기침 악화되었으며, 내원 당일 아침 손가락 2마디 정도의 객혈 보임.

History

- **Past Medical History**

- HTN, DM, DL
- **BE, chronic bronchiolitis (2013)**
- **Pneumonia (2019, 2022, 2023, 2024)**
- **R/O NTM lung disease (2022)**
- PCV 13 vaccination (2019)
- Breast ca. (2002)
- **RA (20년 전)**
- **Lt. malleolar cellulitis with sore wound (2025)**

- **Operation History**

- Rt. Mastectomy (2002)
- Debridement for Lt. ankle cellulitis/sore wound (2026.04.23)

- **Medication History**

- RA: tacrolimus 1mg bid, methylprednisolone 0.5mg qd, celecoxib 200mg bid, diacerein 50mg bid
- Cellulitis: cefaclor 375mg qd, ciprofloxacin 250mg qd → 보름 전 중단
- HTN/DM/DL medication

- **Family History**

- 오빠: 심근경색 (14년 전)

- **Social History**

- **Never-smoker**
- Alcohol Hx. (-)

ROS & Physical Examination

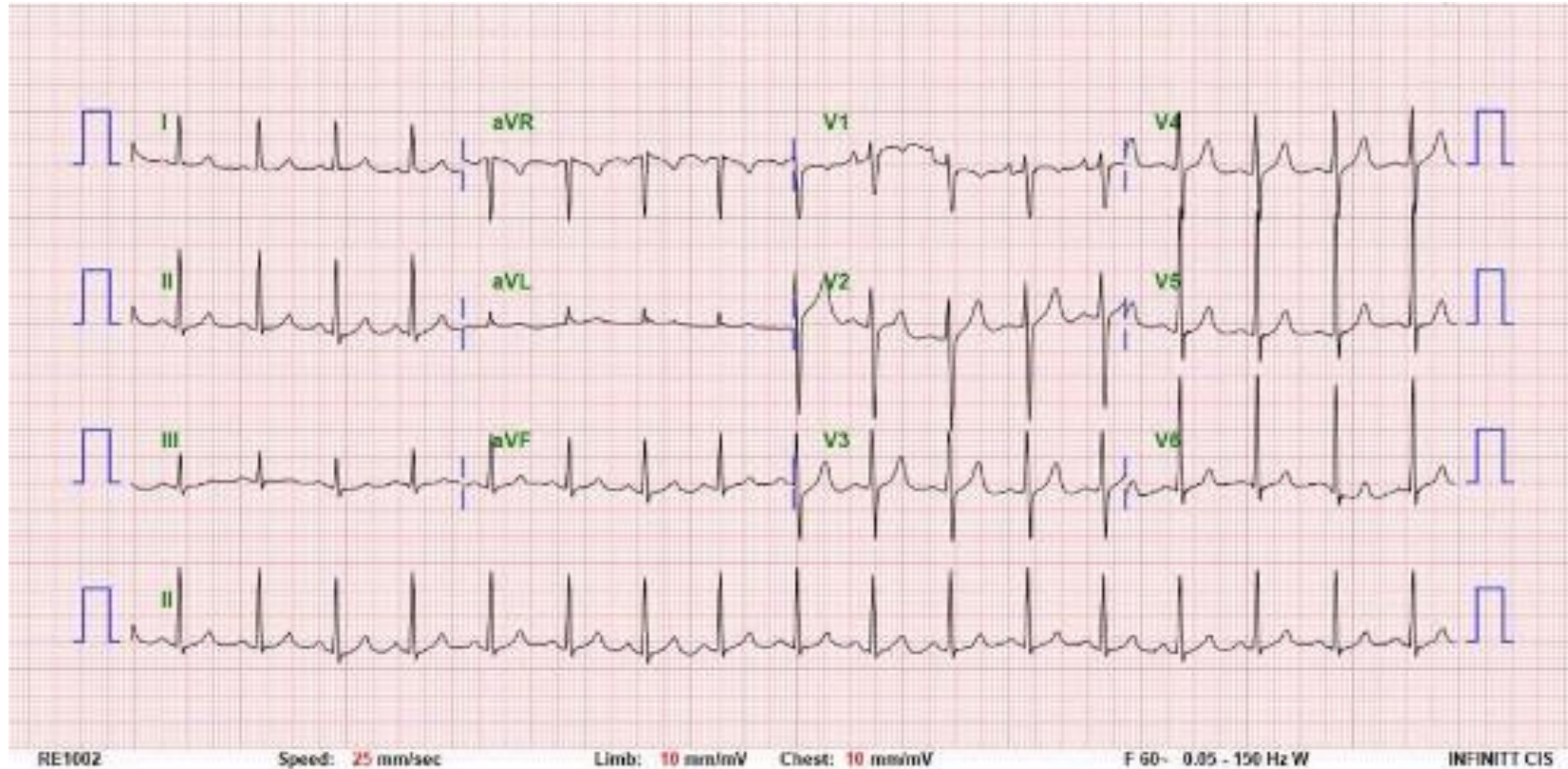
- ROS

General	Weakness (-) / Fever (+) / Chill (-) / Fatigue (-) / Weight gain (-) / Weight loss (-)
HEENT	Headache (-) / Dizziness (+) / Tinnitus (-) / Sore throat (-)
Respiratory	Cough (+) / Sputum (+, 노란색) / Rhinorrhea (+) / Dyspnea (+) / Hemoptysis (+) / Orthopnea (-)
Cardiovascular	Chest pain (-) / Palpitation (-) / Arrhythmia (-)
Gastrointestinal	Anorexia (-) / Nausea (-) / Vomiting (-) / Diarrhea (-) / Constipation (-) / Abdominal pain (-) / Hematemesis (-) / Melena (-) / Hematochezia (-)
Genitourinary	Frequency (-) / Urgency (-) / Nocturia (-) / Dysuria (-) / Hematuria (-)

- Physical Examination

V/S	BP 160/100 mmHg, HR 101/min, RR 19/min, BT 38.0°C, SpO2 100% (ER) Initial SpO2: 92% → 구급차에서 O2 nasal 3L 후 98%
GA/GC	General appearance : Acute Ill looking General condition : Fair Mentality : Alert (GCS 15)
HEENT	Sclera: Anicteric , Conjunctiva: Non anemic , Tongue: Not dehydrated
Chest	Heart sound: Normal heart sound without murmur Breathing sound: 흡기 시 both lower rhonchi
Abdomen	Tenderness/Rebound tenderness : (-/-) , CVAT (-/-)
Extremity	Edema (-)

ECG (2026.05.19)

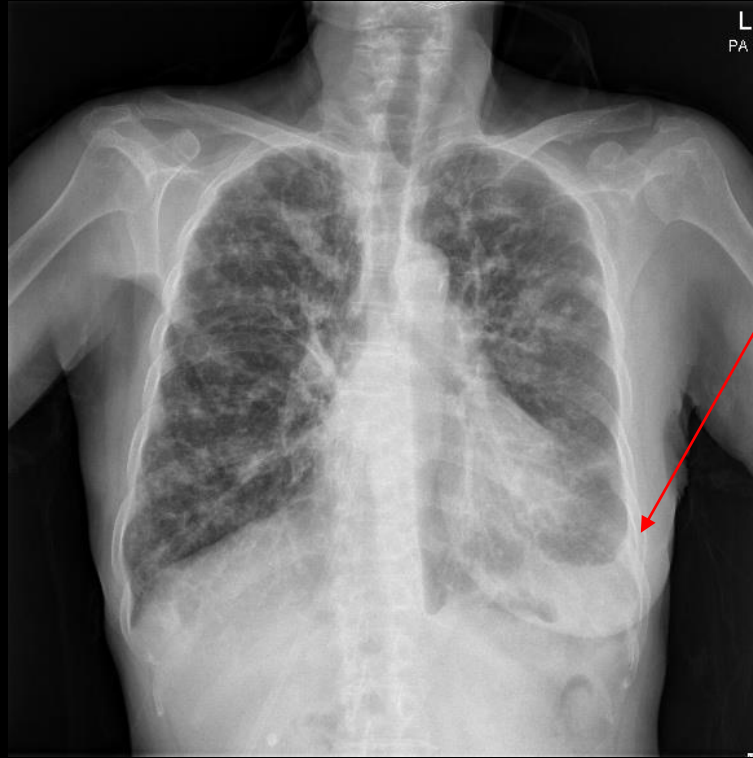


Sinus tachycardia, HR 105

Initial Image findings – CXR



이전 내원 당시 CXR
(2025.09.26)



이번 내원 CXR
(2026.05.19)

- **Aggravated multifocal patchy nodular opacities, both lung**
- Persistent bilateral CP angle blunting, with **increased Lt. side**
- Cardiomegaly

Initial impression

- **Initial impression**
 - R/O Pneumonia (CAP)
 - R/O Acute exacerbation of chronic bronchiolitis/bronchiectasis
 - R/O Aggravation of underlying NTM lung disease
 - R/O Heart failure
 - R/O Pulmonary embolism (PE)

Diagnostic Plan

- **Diagnostic Plan**

- Laboratory test (CBC, CRP, procalcitonin, ABGA, BNP/NT-proBNP, D-dimer)
- Blood, Sputum Gram stain & culture, Sputum AFB stain & culture
- PCR & Antigen Test
 - Sputum M. Tb/Rifampin 신속 PCR, NTM PCR
 - Nasopharyngeal swab (Ag/PCR)
 - Urine Ag test
- Chest CT / CTPA

Laboratory findings (2026.05.19)

CBC

검사명	결과	참고치	단위
WBC	16.47	3.8~11.0	10E3/uL
RBC	2.34	3.5~5.4	10E6/uL
Hb	7.4	11.2~15.0	g/dL
Hct	21.9	34.0~44.3	%
seg neutrophil	82.6	40.0~70.4	%
lymphocyte	7.8	20.0~48.0	%
monocyte	6.1	4~11.5	%
eosinophil	3.2	0~6.0	%
basophil	0.3	0~2	%

Electrolyte & RFT & LFT

검사명	결과	참고치	단위
Na	133.1	138~148	mmol/L
K	5.42	3.5~5.3	mmol/L
Cl	106.9	100~110	mmol/L
BUN	26.5	6~26	mg/dL
Creatinine	1.15	0.4~1.2	mg/dL
AST	24	0~40	U/L
ALT	39	0~40	U/L
ALP	89	40~129	U/L
T. Bilirubin	0.37	0.1~1.2	mg/dL
T. Protein	7.22	6.0~8.0	g/dL
Albumin	3.26	3.3~5.2	g/dL

ABGA

검사명	결과	참고치	단위
pH	7.41	7.35~7.45	
PO2	117	83~108	mmHg
PCO2	25	35~45	mmHg
HCO3	15.8	21~28	mM
Lactic acid	0.7	0.7~2.5	mmol/L

Inflammatory marker

검사명	결과	참고치	단위
CRP	9.4	0~0.5	mg/dL
Procalcitonin	0.31	0.05	ng/mL

HF/PE 관련 marker

검사명	결과	참고치	단위
proBNP	613	0~349	pg/mL
CK-MB	0.53	0.18~5.0	ng/mL
D-dimer	7.94	0~0.5	ug/mL

Laboratory findings (2026.05.19)

Infectious work-up

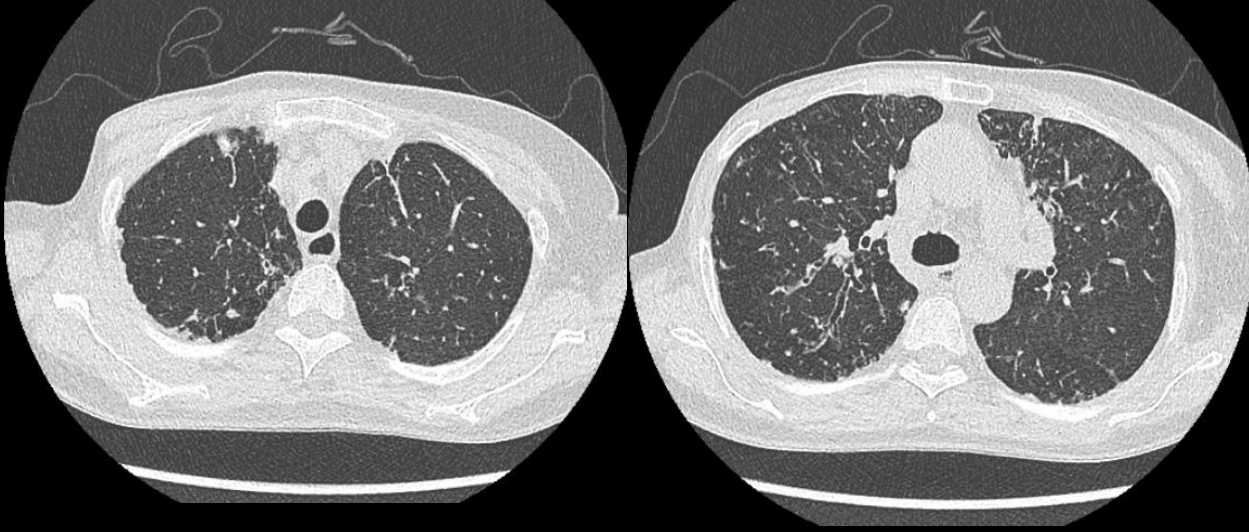
검사명	결과
Nasopharyngeal swab	
Influenza A 항원신속검사	Negative
Influenza B 항원신속검사	Negative
COVID-19 rapid Ag test	Negative
Blood	
HIV 항원/항체 검사	Negative (0.11)
Chlamydia pneumoniae Ab IgM	Negative (3.3)
M. pneumoniae IgM	Negative (0.3)
Urine	
Legionella Ag	Negative
S. pneumoniae Ag	Negative
Sputum	
AFB 농축염색	Negative
AFB culture	Pending
MTB PCR	Not detected
NTM PCR	Not detected
M. Tb/Rifampin 신속 PCR	Not detected

Urinalysis

검사명	결과	참고치	단위
Urine WBC	0~2	0~2	/HPF
Leukocyte esterase	-	-	
Urine nitrite	Negative	Negative	

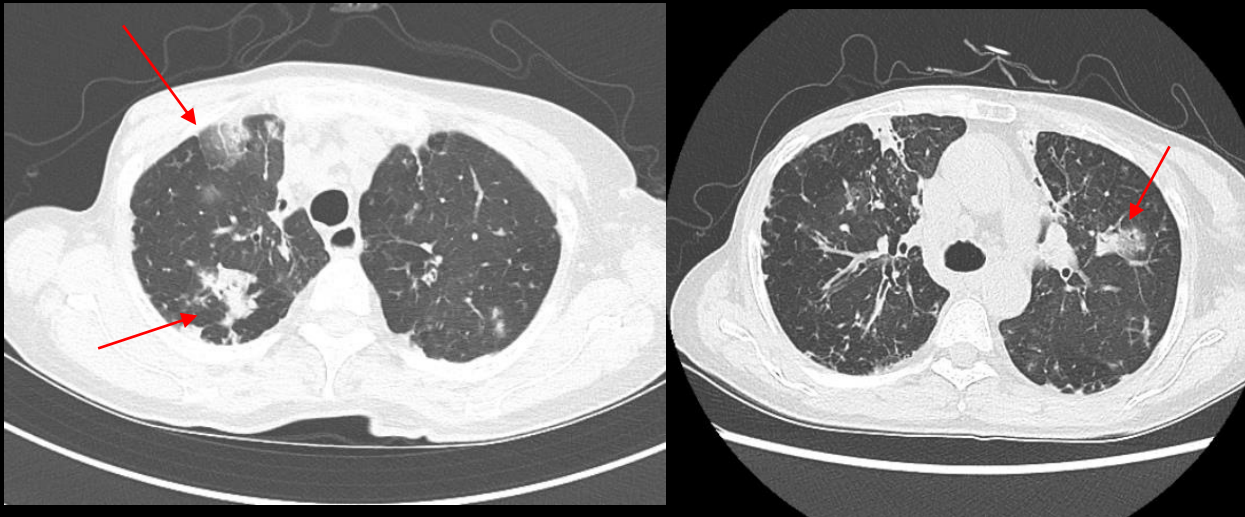
- Culture
 - Blood : No growth, 5 days
 - Sputum : Poor quality, Candida albicans (Few)
 - Urine: 2,000 CFU/mL (VSE, CRE)

Chest CT



2025.09.26

2026.05.19

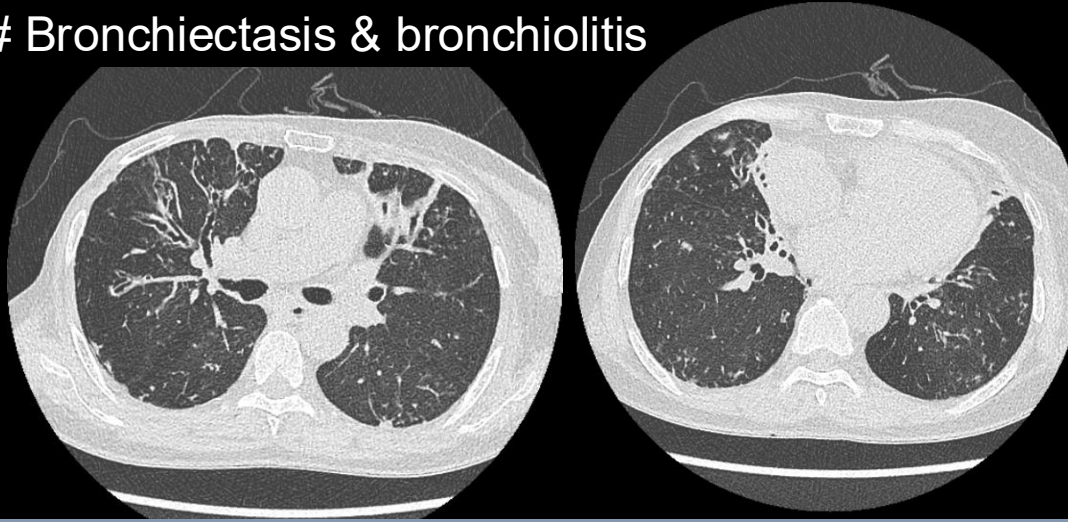


→ Pneumonia,
most likely

- Increased multifocal consolidation & GGO

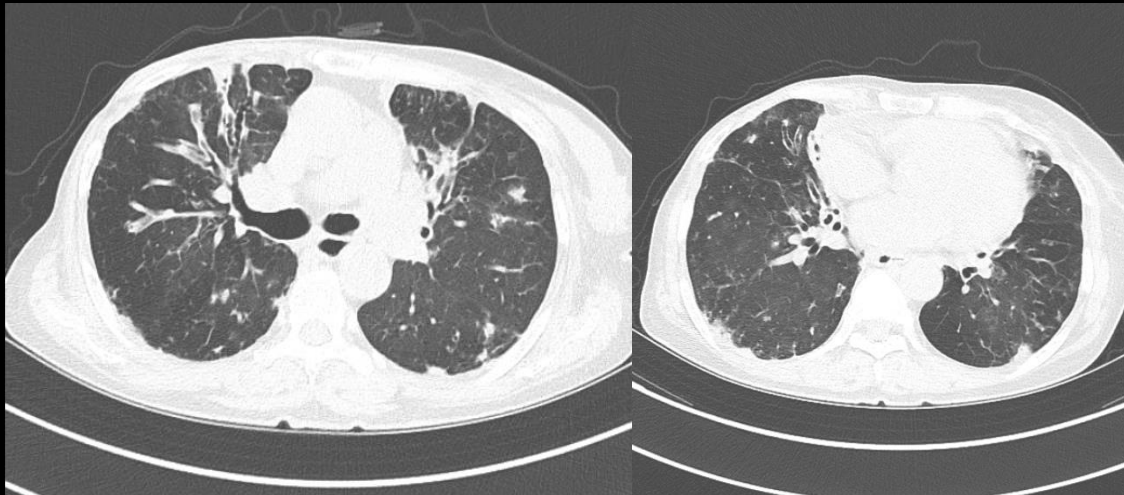
Chest CT

Bronchiectasis & bronchiolitis



2025.09.26

2026.05.19



- No interval change

→ Acute exacerbation of bronchiolitis, less likely

→ Aggravation of underlying NTM lung disease, less likely

Assessment

- Nonsevere CAP
 - 기침 악화, 노란색 가래, 발열, 호흡곤란의 임상 증상
 - New infiltrate on CT Chest (GGO, consolidation)
- R/O NTM lung disease
- Hemoptysis associated with BE

TABLE 131-3 Criteria for Severe Community-Acquired Pneumonia

Minor criteria

- Respiratory rate ≥ 30 breaths/min
- $\text{PaO}_2/\text{FiO}_2$ ratio ≤ 250
- Multilobar infiltrates
- Confusion/disorientation
- Uremia (BUN level ≥ 20 mg/dL)
- Leukopenia (WBC count < 4000 cells/ μL)
- Thrombocytopenia (platelet count $< 100,000$ cells/ μL)
- Hypothermia (core temperature $< 36^\circ\text{C}$)
- Hypotension requiring aggressive fluid resuscitation

Major criteria

- Respiratory failure requiring invasive mechanical ventilation
- Septic shock requiring vasopressors

Abbreviations: BUN, blood urea nitrogen; $\text{PaO}_2/\text{FiO}_2$, arterial oxygen pressure/fraction of inspired oxygen; WBC, white blood cell.

Therapeutic Plan

1. CAP

- 병동 입원 : CURB-65 score 2점

TABLE 131-2 Epidemiologic Factors Suggesting Possible Causes of Community-Acquired Pneumonia

FACTOR	POSSIBLE PATHOGEN(S)
Alcoholism	<i>Streptococcus pneumoniae</i> , oral anaerobes, <i>Klebsiella pneumoniae</i> , <i>Acinetobacter</i> spp., <i>Mycobacterium tuberculosis</i>
COPD and/or smoking	<i>Haemophilus influenzae</i> , <i>Pseudomonas aeruginosa</i> , <i>Legionella</i> spp., <i>S. pneumoniae</i> , <i>Moraxella catarrhalis</i> , <i>Chlamydia pneumoniae</i>
Structural lung disease (e.g., bronchiectasis)	<i>P. aeruginosa</i> , <i>Burkholderia cepacia</i> , <i>Staphylococcus aureus</i>
Dementia, stroke, decreased level of consciousness	Oral anaerobes, gram-negative enteric bacteria

- 경험적 항생제 (Piperacillin/tazobactam 4.5g IV q6h+levofloxacin 750 mg IV q24h)
 - *Pseudomonas* 위험인자 있어 항녹농균 커버
- Tacrolimus, Methylprednisolone 중단

TABLE 131-5 Initial Treatment for Inpatients with or without Risk Factors for Infection with MRSA or *Pseudomonas aeruginosa*

DISEASE SEVERITY, RISK STATUS	REGIMEN
Nonsevere	
No risk factors	A β -lactam ^a + a macrolide ^b or A respiratory fluoroquinolone ^c
Prior respiratory isolation	Add coverage for MRSA ^d or <i>Pseudomonas aeruginosa</i> ^e
Recent hospitalization, antibiotic treatment, $\pm$ LV ^f	Add coverage for MRSA ^d or <i>P. aeruginosa</i> ^e only if cultures are positive

on serum levels) or linezolid (600 mg q12h). ^ePiperacillin-tazobactam (4.5 g q6h), cefepime (2 g q8h), ceftazidime (2 g q8h), imipenem (500 mg q6h), meropenem (1 g q8h), or aztreonam (2 g q8h). ^fObtain cultures. MRSA rapid nasal polymerase chain reaction can also be used if available.

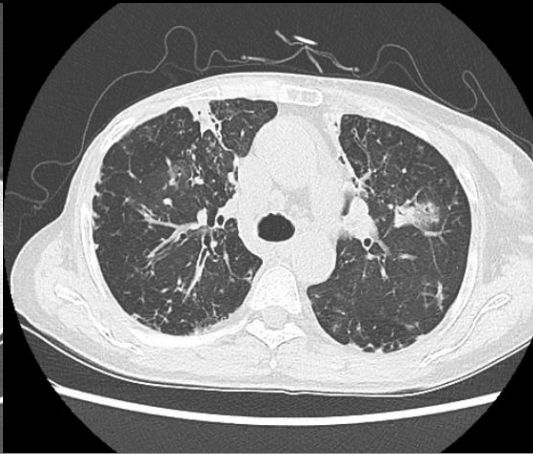
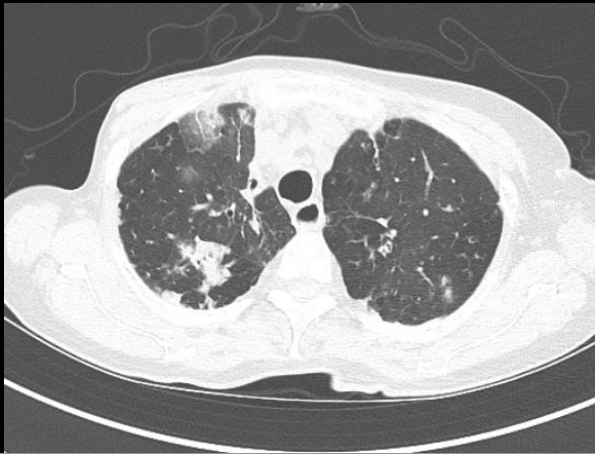
Therapeutic Plan

2. Hemoptysis associated with bronchiectasis

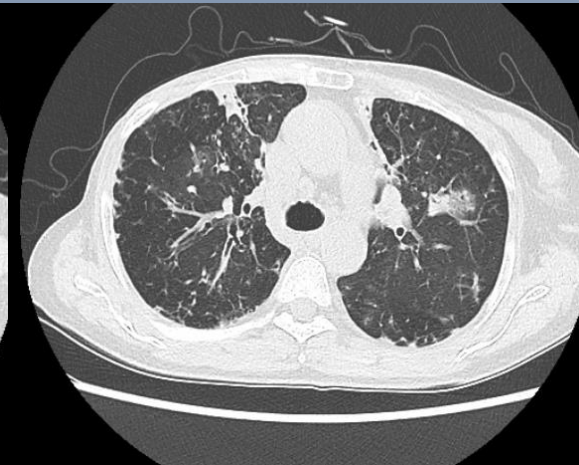
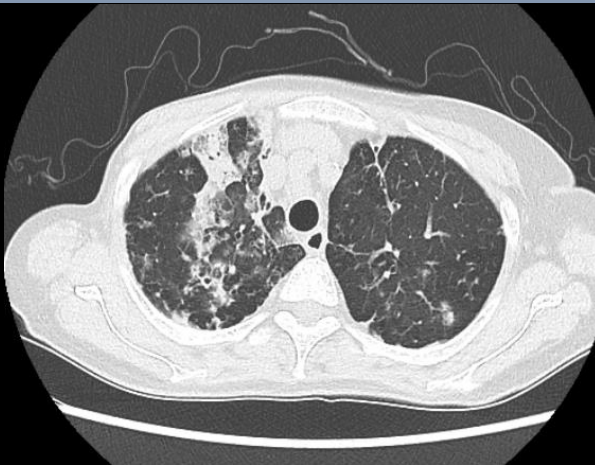
- Tranexamic acid 500mg/5mL 4AMP IV q24h : 출혈량 감소
- Codeine 20mg qod : 기도 자극 및 객혈 악화 방지
- Ambroxol HCl Amp IV q8h : 객담 배출 촉진
- 침상 절대 안정 : 객혈 증가 방지 및 호흡 안정화

Progress Note

	05.22	05.19	
CRP	13.31	9.4	mg/dL



2026.05.19



2026.05.22

CT chest : multifocal GGO 호전 없음
→ 항생제 변경 : meropenem 500mg IV q8h
+ levofloxacin 750mg IV q24h

Progress Note

	05.26	05.22	
CRP	2.29	13.31	mg/dL



2026.05.19



2026.05.26

Haziness 호전

→ 퇴원

Progress Note

- 퇴원 약제

CAP

Levofloxacin Tab 750 mg qd 7일

Cefditoren Pivoxil Tab 100 mg tid 7일

Hemoptysis

Codeine Phosphate 20 mg tid 7일

DL-methylephedrine HCl/Ammonium chloride Etc. 20 ml qid 7일

Disease review

Pneumonia:

Community-Acquired Pneumonia

Pneumonia

Definition

- Infection of the pulmonary parenchyma
- 미생물이 폐포 (alveoli) 내에서 증식하고 이에 대한 숙주의 염증 반응이 발생한 결과

Classification

- Community-acquired (CAP)
- Hospital-acquired (HAP)
- Ventilator-acquired (VAP)

Pathophysiology

Past

Introduction of potential pathogens into a sterile lung

Now

Disruption of lung microbiota homeostasis

- epithelial $\pm$ endothelial injury : viral infection, microaspiration
- Release of cytokines, chemokines, and catecholamines
- Selective bacterial overgrowth : *S. pneumoniae*, *Pseudomonas aeruginosa*
- Inflammation-bacterial growth positive feedback loop
- Clinically apparent pneumonia

Pathology

Lobar pneumonia의 병태 진행 과정

Edema	• Protein-rich exudate • Bacteria in alveoli
Red Hepatization	• Erythrocytes in intra-alveolar exudate
Gray Hepatization	• Lysis and degradation of erythrocytes • Predominant neutrophils • Abundant fibrin deposition • Containment of infection
Resolution	• Predominant macrophages • Clearance of bacteria and inflammatory debris

Etiology of CAP

- Increasing importance of respiratory viruses
- Frequent failure to identify a specific pathogen

Typical Bacteria	• <i>S. pneumoniae</i>, <i>H. influenzae</i> • <i>S. aureus</i>, <i>Klebsiella pneumoniae</i>, <i>Pseudomonas aeruginosa</i> (in selected patients)
Atypical Pathogens	• <i>Mycoplasma pneumoniae</i> • <i>Chlamydia pneumoniae</i> • <i>Legionella</i> spp
Respiratory Virus	• Influenza virus, RSV, Parainfluenza virus

Consideration of MRSA or *P. aeruginosa* in selected high-risk patients

TABLE 131-2 Epidemiologic Factors Suggesting Possible Causes of Community-Acquired Pneumonia

FACTOR	POSSIBLE PATHOGEN(S)
Alcoholism	<i>Streptococcus pneumoniae</i> , oral anaerobes, <i>Klebsiella pneumoniae</i> , <i>Acinetobacter</i> spp., <i>Mycobacterium tuberculosis</i>
COPD and/or smoking	<i>Haemophilus influenzae</i> , <i>Pseudomonas aeruginosa</i> , <i>Legionella</i> spp., <i>S. pneumoniae</i> , <i>Moraxella catarrhalis</i> , <i>Chlamydia pneumoniae</i>
Structural lung disease (e.g., bronchiectasis)	<i>P. aeruginosa</i> , <i>Burkholderia cepacia</i> , <i>Staphylococcus aureus</i>
Dementia, stroke, decreased level of consciousness	Oral anaerobes, gram-negative enteric bacteria
Lung abscess	CA-MRSA, oral anaerobes, endemic fungi, <i>M. tuberculosis</i> , nontuberculous mycobacteria
Travel to Ohio or St. Lawrence river valley	<i>Histoplasma capsulatum</i>
Travel to southwestern United States	Hantavirus, <i>Coccidioides</i> spp.
Travel to Southeast Asia	<i>Burkholderia pseudomallei</i> , avian influenza virus
Stay in hotel or on cruise ship in previous 2 weeks	<i>Legionella</i> spp.
Local influenza activity	Influenza virus, <i>S. pneumoniae</i> , <i>S. aureus</i>
Exposure to infected humans	SARS-CoV-2
Exposure to birds	<i>H. capsulatum</i> , <i>Chlamydia psittaci</i>
Exposure to rabbits	<i>Francisella tularensis</i>
Exposure to sheep, goats, parturient cats	<i>Coxiella burnetii</i>

Abbreviations: CA-MRSA, community-acquired methicillin-resistant *Staphylococcus aureus*; COPD, chronic obstructive pulmonary disease; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.

Clinical Manifestations of CAP

Symptoms (무증상~중증)

- Fever, tachycardia, chills, sweats
- Cough $\pm$ sputum
- Dyspnea
- Pleuritic chest pain
- GI symptoms : nausea, vomiting, diarrhea
- Fatigue, headache, myalgia, arthralgia

- Elderly : confusion, 만성질환 악화
- Necrotizing pneumonia : gross hemoptysis

Physical Examination

- Tachypnea, 부호흡근 사용
- Altered tactile fremitus
- Dullness to percussion
- Crackles, bronchial breath sounds, pleural friction rub

Diagnosis of CAP

Clinical diagnosis

- Compatible symptoms (cough, sputum, fever, dyspnea)
+ New infiltrate on chest radiography

Differential diagnosis

- Acute bronchitis, COPD exacerbation, Heart failure, Pulmonary embolism

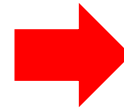
Etiologic diagnosis

- **Specific pathogen identification is often unsuccessful**
- Sputum Gram stain & culture, blood cultures, urinary antigen test, PCR, Inflammatory biomarkers (CRP, PCT)

Severity Assessment

① CURB-65

Confusion
Urea > 7mmol/L
Respiratory rate $\geq$ 30/min
Blood pressure – systolic $\leq$ 90mmHg or diastolic $\geq$ 60mmHg
Age $\geq$ 65 years

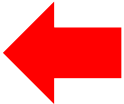


Score 0	Outpatient
Score 1-2	Hospitalization
Scores $\geq$ 3	ICU admission

Severity Assessment

② PSI (Pneumonia Severity Index)

Class 1 : 0 (0.1%)	Outpatient therapy
Class 2 : <70 (0.6%)	
Class 3 : 71-90 (0.9%)	Short hospital admission
Class 4 : 91-130 (9.3%)	Inpatient care
Class 5 :131-405 (27.0%)	



Demographic factors	Male: Age
	Female : Age-10
	Nursing home resident : 10
Comorbid illness	Neoplastic disease (active) : 30
	Chronic liver disease : 20
	Heart failure : 10
	Cerebrovascular disease : 10
	Chronic renal disease : 10
Physical examination findings	Altered mental status : 20
	Respiratory rate≥30/min : 20
	Systolic BP< 90mmHg : 20
	Temperature<35°C or ≥40°C : 15
	Pulse≥125/min : 10
Laboratory and radiographic findings	Arterial pH<7.35 : 30
	BUN≥30mg/dL : 20
	Sodium<130mEg/L : 20
	Glucose≥250mg/dL : 10
	Hematocrit<30% : 10
	PaO2<60mmHg or SaO2<90% : 10
	Pleural effusion : 10

Treatment of CAP

Initial Antibiotics

Outpatients

No comorbidity or no risk factors for antibiotic resistance	• Amoxicillin (1g tid) + either with macrolide or doxycycline (100mg bid)
	• Doxycycline (100mg bid)
	• Macrolide
With comorbidity ± risk factors for antibiotic resistance	• Amoxicillin/clavulanate or cephalosporin + either with macrolide or doxycycline (100mg bid)
	• Respiratory fluoroquinolone

Inpatients

Nonsevere (Non-ICU)	
No risk factors	• β-lactam + macrolide or Respiratory fluoroquinolone
Prior respiratory isolation	• + Coverage for MRSA or Pseudomonas aeruginosa
Recent hospitalization, antibiotic treatment, ± local validation	• + Coverage for MRSA or Pseudomonas aeruginosa (only when positive cultures)
Severe (ICU)	
No risk factors	• β-lactam + macrolide or β-lactam + respiratory fluoroquinolone
Prior respiratory isolation, recent hospitalization, antibiotic treatment, ± local validation	• + Coverage for MRSA or Pseudomonas aeruginosa

Treatment of CAP

Adjuvant Treatment

- Oxygen therapy
- Hydration
- Vasopressor treatment (for septic shock)
- Assisted ventilation (if needed)
- Glucocorticoid (selected severe CAP)

Failure to improve

Reevaluation

- No clinical improvement by Day 3
- Or earlier if clinical deterioration

Possible causes

- Wrong diagnosis, unexpected or drug-resistant pathogen, sequestered focus (lung abscess, empyema), inadequate dose or dosing interval, nosocomial superinfection

Further evaluations

- Chest CT or bronchoscopy

Complications & Follow-up

Complications of severe CAP

- Respiratory failure
- Septic shock / multiorgan failure
- Lung abscess
- Complicated pleural effusion
- Cardiovascular events (HF, arrhythmia, MI, stroke)

Follow-up

- Inpatients : 4~6주 후 CXR
- Outpatients : 5~7일 내 증상 호전된 비흡연자의 경우 routine CXR 불필요

Reference

- Harrison's Principles of Internal Medicine (22nd Ed.)
- Calculator: Community-Acquired Pneumonia Severity Index (PSI) for Adults. UpToDate.
- Community-acquired pneumonia in Adults: Assessing severity and determining the appropriate site of care. UpToDate.